

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for SCADA System

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R01	14.07.22	Issued for Review
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R04	02.11.22	Reissued for Review

Revision Philosophy:

- All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-012

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

1. INTRODUCTION	6
2. REGULATIONS, CODES AND STANDARDS	8
3. SCADA SYSTEM	11
4. PROGRAMMABLE LOGIC CONTROLLERS FOR ESD	14
5. RTU ROOM AND COMMUNICATION	17
6. EARTHING	18
7. SCADA SIZING BASIS	19
8. INSPECTION AND TESTING REQUIREMENT	20
9. TRAINING	21
10. PACKING	22
11. WARRANTY	23
12. VENDOR DATA REQUIREMENT SCHEDULE (VRDS)	24

Abbreviations

IEC	International Electro technical Commission
ISA	Instrument Society of America
ITU	International telecommunication Union
SCADA	Supervisory Control and Data Acquisition System
FAT	Factory Acceptance Test
SAT	Site Acceptance Test
QAP	Quality Assurance Plan
PWAMC	Post Warranty Annual Maintenance Contract

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

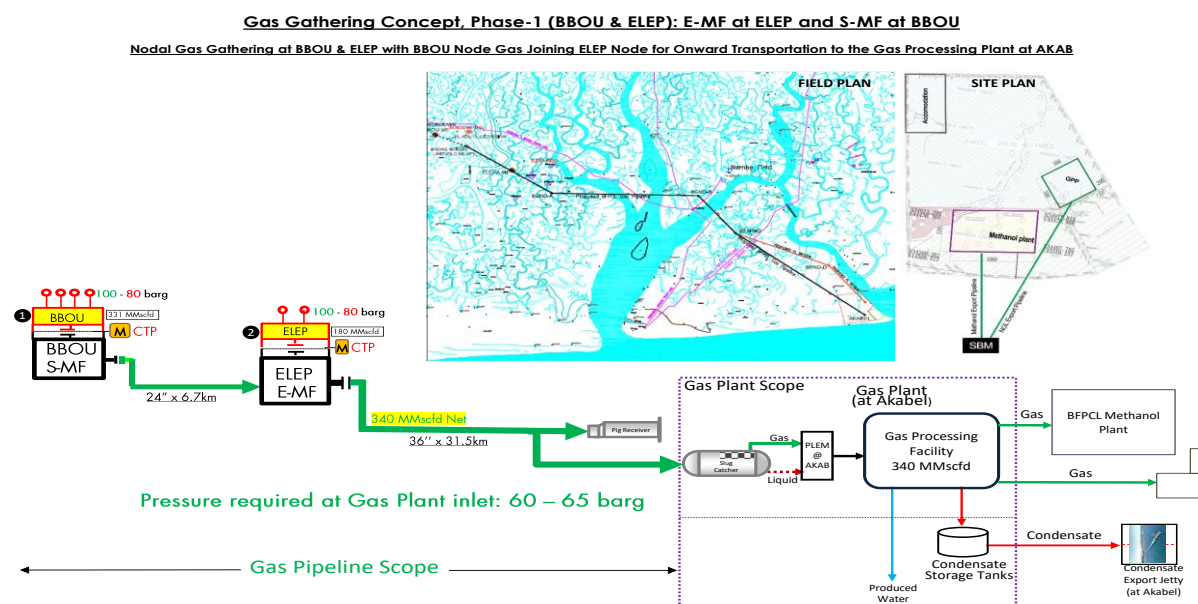


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document defines the basic requirements for the SCADA Systems for BBOU,ELEP AND AKAB. The FEED shall fully comply with all specified relevant contractual requirements. This document defines the basic requirements for the engineering and design of the instrumentation for BBOU, ELEP and AKAB pipeline. The purpose of this specification is to ensure a standard approach is maintained to the engineering and design of instrumentation and equipment and that it complies with all relevant national and international standards.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

The following standards and codes of practices in it's latest revision shall used for the design,material,selection and installation of the system :

Sr. No.	Standard	Description
1.	IEC-61508	Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems (E/E/PE, or E/E/PES).
2.	IEC-61511	Functional safety - Safety instrumented systems for the process industry sector.
3.	DIN V19250	Control technology; fundamental safety aspects to be considered for measurement and control equipment.
4.	IEEE 802.3	A set of standard specifications for Ethernet Communication
5.	OPC UA	OPC Unified Architecture (OPC UA) is a cross-platform, open-source, IEC62541 standard for data exchange from sensors to cloud applications developed by the OPC Foundation
6.	IEC 61000	Electromagnetic Compatibility
7.	IEC 62831	Automation systems in the process industry - Factory acceptance test (FAT), site acceptance test (SAT), and site integration test (SIT).
8.	API RP 552	Transmission Systems
9.	API RP 554	Process Instrumentation and Control
10.	API RP557	Guide to advanced process control
11.	ISA S50.02	Fieldbus Standard For Use In Industrial Control Systems Part 2: Physical Layer Specification And Service Definition.
12.	ISA S5.2	Binary Logic Diagrams For Process Operations.
13.	ISA S5.3	Graphic Symbols For Distributed Control/Shared Display Instrumentation, Logic And Computer Systems
14.	ISA S5.4	Instrument Loop Diagrams.
15.	ISA S71.04	Environmental Conditions For Process Measurement And Control Systems: Airborne Contaminants.
16.	IEC-60529	Degrees of protection provided by enclosures (IP Code).
17.	IEC-60079	Explosive Atmospheric Standards
18.	IEC 61131	International standard for programmable controller programming languages
29.	ISA S84.01	Application of Safety Instrumented Systems for the Process Industries.

Sr. No.	Standard	Description
20.	IEC-81345-2	Industrial systems, installations and equipment and industrial products - Structuring principles and reference designations - Part 2: Classification of objects and codes for classes.
21.	IEC-82079-1	Preparation of information for use (instructions for use) of products — Part 1: Principles and general requirements.
22.	ISO-9001	Quality Management

3. SCADA System

- 3.1. Each installation (BBOU, EEP and AKAB) shall have a SCADA System (SIL-3 PLC) that shall gather the data from the field instrument for monitoring, shutdown and communication. The SCADA shall have functional requirement of :
- PLC - Provides primary process control, monitoring, and data acquisition functions for the plant.
 - SIS/ESD - Provides protection of personnel, environment, and equipment through detection and actuation of devices designed to lead the plant into a safe condition.
 - F&G - Provides local and centralized warning of personnel to hand fire and gas events, through detection of gas leakage or fire.
 - 2003 logic pressure logic shall be used for the shutdown of the flowline and opening of the corresponding blowdown valve. Voting Logic shall be implemented for Gas release and fire /flame signals
 - It shall interface the CCTV system
 - It shall interface with the Satellite Communication System.
- 3.2. It shall consist of plant I/O interface unit.
- 3.3. The scan time shall be 250 msec for close loop controls and 500 msec for open loops. For the shutdown, it shall have typically 100 ms scan time. Controller shall not be loaded more than 60% of its capacity in terms of I/O and memory, whichever is lower.
- 3.4. For I/O cards number of input/output channels shall be as per manufacturer standard.
It shall comprise of the following
- Analogue input signals
 - Digital input signals
 - Digital output signals
 - Serial Interfaces
- I/O cards should NOT more than 16 channels for AI/AO, 32 channels for DI/DO.
- 3.4.1. The analog input cards shall accept 4~20 mA DC two wire or four wire signals, 1~5VDC signals. Two wire transmitters shall be powered from the system. All digital outputs and digital inputs shall be provided with protecting fuse terminals, all analog inputs and analog outputs shall be provided with knife edge terminals in cabinet of PLC.
- 3.4.2. Digital input cards shall be 24VDC .All digital inputs are potential free contacts, card shall provide interrogation voltage to the input. These cards shall have channel LED for status indication.
- 3.4.3. Digital output cards shall be 24V DC. Interposing relay shall be used. Interposing relays shall be solid state type. Electromechanical relays are not acceptable.
- 3.4.4. Surge Protection Devices (SPDs) shall be used for every I/O points. Barriers shall be provided for IS signals from hazardous areas.

3.5. Engineering and Operation Station

A panel mounted engineering cum operation interface shall be provided with Emergency pushbuttons.

3.6. System Architecture and Database

3.6.1. System Architecture shall be composed of standards-based technology including latest version of approved Windows operating system with a minimum support for 10 years from the date of commissioning / hand-over to owner from Micro-soft (If Microsoft support for OS (Operating system installed for PLC is less than 10 years after hand-over date the first upgrade for OS shall be FREE OF COST to owner when support for installed OS get over (not available) from Microsoft.).

3.6.2. PLC shall comply to Open System Architecture and all communication between workstations and CPU shall be on high-speed Ethernet TCP/IP protocol according to IEEE standards.

3.6.3. The SCADA system shall maintain a database which shall allow the operator to view the information directly and indirectly related to the plant being monitored and controlled. The control system shall handle data which shall reflect:

- general alarms, group status alarms and shutdown alarms and pre-alarms.
- setpoints ,measured variable and control elements status/position of all control loops
- equipment status (e.g. running , stopped ,tripped,off-load,isolated for maintenance or start-up etc)
- archiving, trending and reporting, diagnostic functions.

3.7. System Integrity

The SCADA system is an essential part of the plant operability and safety. SIL-3 certified system shall be used. It shall maintain a high availability with high level of integrity. This shall be achieved by using high reliability components, hardware redundancy where appropriate and good diagnostic capability for trouble shooting.

3.8. Availability and Reliability

3.8.1. The control system shall have a high degree of tolerance to malfunction of hardware, software and operator miskeying .Any fault developed shall have its effect to localized to that part of the control system which remains operational, with a possible reduction in level of functionality. The availability of the control system for control and process monitoring shall be in excess of 99.95%.

3.8.2. Reliability data and minimum mean time between failures (MTBF) of all equipment shall be provided by OEM of PLC during detail engineering. The vendor shall provide MTBF - calculated for 50 degC cabinet internal temperature. Mean time to repair (MTTR), explains the basis of calculation, and provides reliability analysis description and relevant data. The cumulative fault repair time of the whole system shall not exceed eight hours. Moreover, the design, assembly and backup of the system shall prevent the shutdown of the whole system under any foreseeable circumstances. The Seller shall provide the actual operation availability of similar devices ($A = \frac{MTBF}{MTBF + MTTR}$)

3.9. Generation of Report

Report generation facility shall be provided for automatic and on-demand reports. All

historical data shall be accessible for report generation. Automatic Reports: It shall be possible to use historical data points and generate automatically shift (8 hours), daily (24 hours), weekly and monthly reports for the purpose of plant control and plant process information. Reports shall be used for plant control contains listing of critical measurements and compressed data. It shall be possible to produce special reports for analysis purpose by operator or plant engineer using historical and compressed data points on demand. All report start shall have time stamp with dates/times.

4. Programmable logic controllers for ESD

- 4.1. PLC's shall be SIL-3 Fail-safe & microprocessor based which shall be used for plant Emergency shutdown.
- 4.2. The system shall allow dependable and effective control of the plant and shall be designed for maximum integrity and reliability. Integrity shall be maintained by providing a fail-safe system. The system shall be TUV approved.
- 4.3. The system shall have facilities to replace failed equipment such as processors, I/O cards, microprocessor, and batteries on-line without a shutdown. RAM memories shall have the necessary built-in backup battery.
- 4.4. The system shall have high integrity including automatic crosschecking, integral self-test facility of input circuits, output circuits, and processors and provide failure indication for operator information and action.
- 4.5. The system shutdown shall only occur in the event of an emergency situation and not as a result of single point failure within the system. A single system fault shall not prevent correct response to a shutdown demand. All components for the controller unit shall be included in the scope of supply (system cabinets, controllers, I/O cards, system cables etc.). The number of processors and ESD I/O modules, cabinets and other accessories shall be defined and indicated in the proposal by the Vendor.
- 4.6. The system shall be modular in construction. The types of modules shall be kept to as minimum as possible in order to have interchangeability and low spares inventory. The system shall have extensive set of self-diagnostics for hardware and software for easy maintenance.
- 4.7. The system shall be immune to noise and shall ensure safe and reliable operation when subjected to RFI and electromagnetic disturbances.
- 4.8. The system shall be able to operate satisfactorily from 150C to 300C and 20% to 80% non-condensing humidity unless otherwise specified.
- 4.9. Operation of the system shall be completely unaffected if there is a momentary powerless of the order of 20 milliseconds, unless otherwise specified.
- 4.10. The system shall be supplied with necessary programming tools and related accessories.
- 4.11. The following items should be redundant:
 - Controller
 - Memory
 - Power supply modules/units mounted in CPU and I/O rack and mounted in system cabinet
 - Communication modules required for communication: Between controllers, Between controllers and HMI, between Controllers & I/O module sub system as possible
 - Communication network cables and accessories.
 - Servers, wherever applicable
 - I/Os for critical loops / signals (to be decided during detailed engineering based

on process and OEM inputs)

- The following guidelines shall be followed during IO grouping:
- No two identical / similar equipments shall be grouped in the same I/O module.
- I/Os related to equipment and I/Os related to its associated auxiliaries shall be connected to different modules.
- Inputs and outputs shall not be combined in a single module. - Where ever 2oo3 logic is involved, the redundant inputs shall be connected to different modules.
- I/O modules of one equipment and associated auxiliaries to be located in the same controller.

- 4.12. The PLC shall be fault-tolerant and built up in modules of standard components and software of the latest proven types and versions and shall have memory protection against power failure. Control room will be air-conditioned using split AC system. The PLC system shall have Field proven system and shall have been in use for at least 2 (Two) Years for the model offered and in similar type of application. PLC Supplier shall not offer the series of modules which will become obsolete in the next 15 years. Also, bidder shall provide support for spares & service for the plant life of 25 years from the date of handover.
- 4.13. Necessary REDUNDENT Cooling Fans for PLC cabinets & Air Filters with Temperature monitoring shall be provided. High Temperature & Fan Failure Alarms shall be provided by PLC vendor. The internal cabinet Temperature shall be maintained below 30 degC. Smoke and Heat detectors shall be installed inside the RTU and shall be interfaced with PLC system.
- 4.14. All I/O connected to control system shall have galvanic/optical isolation depending on applications. All electronic modules of PLCs shall have G3 corrosion protection as per ISA S 71.04
- 4.15. The system shall be designed for the highest safety and availability and all hardware, software and programming shall comply with IEC61508 and IEC 61511
- 4.16. ESD supplier shall offer license for tags considering the I/O count & count of soft tags provided for the plant. 20% spare tags shall be available after commissioning of the system. Spare tags shall be checked during SAT. ESD supplier shall indicate his licensing philosophy in the offer.
- 4.17. The redundancy shall not be counted as part of the spare capacity. The ESD shall provide the following spare capacity after factory acceptance test.
- 40% system spare for additional software application as well as bus communication.
 - 20% free space shall be available in cabinets for addition of hardware (Also include 20% spare interposing relays, terminals, fuses, MCBs).
 - All I/O connected to control system shall have galvanic/optical isolation depending on applications. The design shall consider at least 20% spare channels available in each I/O module and 20% spare slot for each type of module for future expansion

purpose.

- CPU load shall not exceed 60% under normal and abnormal process conditions.

4.18. Following independency requirements shall be considered for system:

- Sensor independent
- Controller independent
- Barriers and marshalling independent
- Junction boxes independent

4.19. All changes in logic status shall be recorded, including change of status, inputs, inactivation of inputs of Interlock-group, first-out alarm, etc. All changes of analog trip values made from the Engineering Work Station shall also be logged. Any inactivation for operational or maintenance purposes shall be logged. All printout messages shall include time (day/hr./min./sec./msec.) and tag numbers. Time resolution shall be as per ESD PLC CPU cycle time. The SOE time stamp resolution shall be 1 millisecond minimum.

5. RTU Room and Communication

- 5.1. SCADA system shall be located in the RTU Room. The RTU room shall be located in each platform (BBOU,ELEP). For AKAB, it shall be located in the main control room.
- 5.2. There will be redundant Fibre Optics Links between BBOU-AKAB and ELEP-AKAB. In the failure of the fibre link, a satellite based telecommunication system shall be automatically turned on.
- 5.3. The satellite based communication shall follow ITU standard.
- 5.4. The data from SPDC well shall be interfaced with the respective platform, namely BBOU and ELEP.

6. Earthing

6.1. The following earth systems shall be provided when applicable

- Protective Earth / Electrical Earth-
- System Earth/ Signal Earth- Control Cabinet Earth (Dirty Earth) – For Signal earths, screens etc.

6.2. **Protective Earth :**

Earth metallic enclosure / cabinet / console etc. shall be provided with electrical earth lug, as a minimum. Unless recommended otherwise by vendor, all earthing lugs of metallic equipment shall be connected individually to electrical protective earthing system bus-bar / earthing station using a maximum of 10sq mm solid copper conductor PVC insulated wires. Where multiple cabinets are multiplexed together, earth looping with permanent shorting link cables shall be acceptable.

6.3. **System Earth :**

6.3.1. System earth shall be totally noise free dedicated earthing system and shall be fully isolated from electrical protective earth. This earth must be very high integrity system and shall be used to ground zero volt references and signal cable grounds.

6.3.2. System earth shall be less than one (1) ohm grounding system recommendation with its own dedicated earth pits. The earth pits shall be suitably located outside the control room and away from any heavy noise plant equipment.

6.3.3. The earth pit design shall be as per international code of practice for earthing. A minimum of two (2) number of earth pits, in redundant configuration, shall be provided for System earth. In case number of pits required to meet system earth resistance are more than one (1) number, the number of system earth pits shall be two times the actual number of pits required to meet the redundancy requirement specified above. All these system earth pits shall be securely connected with each other to form a one homogeneous system earth grid.

6.3.4. Each marshalling / system cabinet etc. shall be provided with system earth bus-bar which shall be insulated from the metallic body frame. This bus-bar shall be used to earth also signal zero volt references and signal cable screens. Terminals used for termination of spare conductor pairs / cores of multi-pair signal / control cables shall be connected to system earth bus-bar. Shorting links shall be used for spare terminal looping.

6.3.5. System bus-bars in the multiplexed cabinets can be joined together by permanent shorting links. System bus-bars of other cabinets can also be connected together provided they are permanently joined using 35 sq. mm stranded copper conductor cable.

6.3.6. The redundant system earth pits shall be connected to the Electrical protective earthing system through suitable surge isolation and protection device like (Isolating Spark Gap- ISG) into equipotential bonding. These lightning surge isolation and protection devices shall comply to the requirements of IEC 62561-3.

7. SCADA Sizing Basis

Attached I/O List for SCADA.

8. Inspection and Testing Requirement

- 8.1. SCADA system shall be inspected and tested by the SPDC/ EPC/Engineering Consultant. The Vendor shall submit to and obtain approval of FAT and SAT procedure at least six weeks prior to the performance of any such test. The FAT/SAT may be witnessed by SPDC/EPC/Engineering Consultant. The purpose of this FAT and SAT is to demonstrate with a total system functional test, that the operation of the control, monitoring, and logic systems is in accordance with the requisition requirements. The system functional test shall include all communication links and the Vendor shall provide the necessary equipment to demonstrate the full functionality of the communication links. Vendor shall perform all necessary ongoing inspection activities in accordance with the approved Vendor Quality Assurance Plan (QAP).

9. Training

- 9.1. SCADA vendor shall arrange training for end users. Different types of courses shall be provided for operation/process engineers and operators, instruments maintenance engineers and technicians. Apart from the hardware and software maintenance training and site training, the vendor shall also provide operation training meant for operation personnel. The various facilities available in the system for operation, actions required during emergencies and identifying the various diagnostics messages shall be the main contents of the operation training courses. The vendors shall clearly quantify the training in their offer.

10. Packing

- 10.1. All the material used for packing, wrapping, sealers, moisture resistant barriers and corrosion preventers shall be recognised brands and shall conform to the best standards in the areas for the articles which is packaged. Sea worthy packing shall be used. Workmanship shall be in accordance with the best commercial practice with the requirement of applicable specifications. There shall be no defects, imperfections or omissions which would tend to impair the protection offered by the package a whole. The package shall be suitable for storing in tropicalized climate. Shipment shall be thoroughly checked for completeness before final packing and shipment.

11. Warranty

- 11.1. The vendor shall be fully responsible for the manufacturing in respect of proper design, quality, workmanship and operation of all equipment, accessories etc. supplied by them for the complement SCADA system for a period as per the commercial terms and conditions of the contract. It shall be obligatory on the part of the vendor to modify and /or replace any hardware and modify the operation applications and software free of cost, in case of any malfunction is revealed even during operation after taking over by the end user within the warranty period. During the warranty period the vendor shall visit the site as per the terms of contract. In the event of any malfunction of the system during warranty period, the vendor shall visit to site with necessary spares within 48 hours of receipt of the complaints.
- 11.2. The vendor shall duly sign the logistic support certificates for the availability of the back-up engineering, maintenance support and spare supports for a period of ten(10) years from the date of expiry of the warranty are closed as part of the contract
- 11.3. Vendor shall quote Post Warranty Annual Maintenance Contract (PWAMC) for five(5) years. PWAMC shall be comprehensive type with all material supply shall be in the scope of vendor.

12. Vendor Data Requirement Schedule (VRDS)

Sr.No	Description	With Pro- posals (prints)	After Purchase Order		Final documenta- tion/ As-built
			Prints	Data needed	
1	Vendor Drawing and Document Schedule		X		X
2	Cabinet/Rack /Console – GA		X		X
3	Cabinet/Rack /Console – Overall Dimensional drgs		X		X
4	Power load & UPS requirements		X		X
5	Heat load requirements		X		X
6	Logging, Alarm, Event & log report format		X		X
7	Bill of Material		X		X
8	Functional Design Specification	X	X		X
9	Factory Testing & Acceptance procedure				X
10	Site Testing & Acceptance procedure		X		X
11	Processor/Bus/Sub system loading Calculation		X		X
12	Catalogue	X	X		X
13	Application/System Software documentation		X		X
14	Hardware specification including Bought out components	X	X		X
15	Training Course Material		X		X
16	Installation, Operation & Maintenance Manuals		X		X
17	Two years of operational spare				
18	Mandatory Spare	X	X		X
19	Consumable spares, Special test tools	X	X		X
20	Proven Track Record		X		X
21	Compliance Statement and deviation list (if any)				
22	Dully Filled in Logistic support Certificate				
23	Soft copy (CD/DVD/Flash Drive) of final doc- umentation				X

Instrumentation Point Database-BBOU (Revision- R03)

Sr. No.	Tag No.	Application	Location	Type of Instruments	IO Type
1	BBOU-FE/FT-1001	Check Meter at the inlet manifold	Field	Venturi Meter	-
2	BBOU-DPT-1001	Differential Pressure Measurement at the inlet across Ve	Field	DP Transmitter	AI
3	BBOU-FI-1001A	Flow Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
4	BBOU-FI-1001B	Flow Indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
5	BBOU-FQI-1001A	Flow Totalizer at RTU	PLC-RTU	On PLC-RTU Screen	Soft
6	BBOU-FQI-1001B	Flow Totalizer at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
7	BBOU-FE/FT-1002	Check Meter at the inlet manifold	Field	Venturi Meter	-
8	BBOU-DPT-1002	Differential Pressure Measurement at the inlet across Ve	Field	DP Transmitter	AI
9	BBOU-FI-1002A	Flow Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
10	BBOU-FI-1002B	Flow Indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
11	BBOU-FQI-1002A	Flow Totalizer at RTU	PLC-RTU	On PLC-RTU Screen	Soft
12	BBOU-FQI-1002B	Flow Totalizer at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
13	BBOU-PI-1001	Battery Limit Pressure at the Inlet Manifold of BBOU	Field	Pressure Gauge	-
14	BBOU-PIT-1001	Battery Limit Pressure at the Inlet Manifold of BBOU	Field	Pressure Transmitter	AI
15	BBOU-PI-1001A	Battery Limit Pressure indication at RTU-Inlet Manifold d	PLC-RTU	On PLC-RTU Screen	Soft
16	BBOU-PI-1001B	Battery Limit Pressure indication at remote DCS-Inlet Ma	Remote DCS	On Remote DCS Screen	V-SAT/Serial
17	BBOU-PAL-1001	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
18	BBOU-PAH-1001	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
19	BBOU-PAL-1001A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
20	BBOU-PAH-1001A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
21	BBOU-TW/TI-1001	Battery Limit Temperature at the Inlet Manifold of BBOU	Field	Temperature Gauge with Thermowell	-
22	BBOU-TW/TE-1001	Battery Limit Temperature at the Inlet Manifold of BBOU	Field	RTD with Thermowell	-
23	BBOU-TIT-1001	Battery Limit Temperature at the Inlet Manifold of BBOU	Field	Temperature Transmitter	AI
24	BBOU-TI-1001A	Battery Limit Temperature at RTU-Inlet Manifold of BBO	PLC-RTU	On PLC-RTU Screen	Soft
25	BBOU-TI-1001B	Battery Limit Temperature at remote DCS-Inlet Manifold	DCS	On Remote DCS Screen	V-SAT/Serial
26	BBOU-PCV-1001	Inlet pressure reduction-BBOU Gathering Manifold	Field	Pressure Regulator	-
27	BBOU-SDV-1001	Inlet Shutdown Valve-BBOU Gathering Manifold	Field	Shutdown Valve- Gas Over Liquid	-
28	BBOU-SD-1001	ESD Command from RTU	PLC-RTU	Push Button	DI
29	BBOU-SD-1001A	ESD Command from Remote DCS	DCS	Push Button	V-SAT/Serial
30	BBOU-SOV-1001A	Inlet SDV Close Signal	Field	Solenoid Valve	DO
31	BBOU-SOV-1001B	Inlet SDV Open Signal	Field	Solenoid Valve	DO
32	BBOU-ZSC-1001	Inlet SDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
33	BBOU-ZSO-1001	Inlet SDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
34	BBOU-ZIC-1001	Inlet SDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
35	BBOU-ZIO-1001	Inlet SDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
36	BBOU-ZIC-1001A	Inlet SDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
37	BBOU-ZIO-1001A	Inlet SDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
38	BBOU-PI-1002	Flowline Pressure to BBOU-A-200 Launcher	Field	Pressure Gauge	-
39	BBOU-PIT-1002	Flowline Pressure for interlock-BBOU Gathering Manifold	Field	Pressure Transmitter	AI

40	BBOU-PI-1002A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
41	BBOU-PI-1002B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
42	BBOU-PAL-1002	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
43	BBOU-PAH-1002	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
44	BBOU-PAL-1002A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
45	BBOU-PAH-1002A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
46	BBOU-PIT-1003	Flowline Pressure for interlock-BBOU Gathering Manifold	Field	Pressure Transmitter	AI
47	BBOU-PI-1003A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
48	BBOU-PI-1003B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
49	BBOU-PAL-1003	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
50	BBOU-PAH-1003	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
51	BBOU-PAL-1003A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
52	BBOU-PAH-1003A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
53	BBOU-PIT-1004	Flowline Pressure for interlock-BBOU Gathering Manifold	Field	Pressure Transmitter	AI
54	BBOU-PI-1004A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
55	BBOU-PI-1004B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
56	BBOU-PAL-1004	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
57	BBOU-PAH-1004	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
58	BBOU-PAL-1004A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
59	BBOU-PAH-1004A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
60	BBOU-BDV-1002	Blowdown Valve to Vent	Field	Blowdown Valve- Gas Over Liquid	-
61	BBOU-SD-1002	ESD Command from RTU	PLC-RTU	Push Button	DI
62	BBOU-SD-1002A	ESD Command from Remote DCS	DCS	Push Button	V-SAT/Serial
63	BBOU-SOV-1002A	BDV Close Signal	Field	Solenoid Valve	DO
64	BBOU-SOV-1002B	BDV Open Signal	Field	Solenoid Valve	DO
65	BBOU-ZSC-1002	BDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
66	BBOU-ZSO-1002	BDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
67	BBOU-ZIC-1002	BDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
68	BBOU-ZIO-1002	BDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
69	BBOU-ZIC-1002A	BDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
70	BBOU-ZIO-1002A	BDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
71	BBOU-PIT-1005	Instrument Gas Manifold Pressure Measurement	Field	Pressure Transmitter	AI
72	BBOU-PI-1005A	Instrument Gas Manifold Pressure Measurement- RTU Indication	PLC-RTU	On PLC-RTU Screen	Soft
73	BBOU-PI-1005B	Instrument Gas Manifold Pressure Measurement- Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
74	BBOU-RO-1001	Restriction Orifice to Vent-BBOU Vent Line Depressurisation	Field	Restriction Orifice	-
75	BBOU-PSV-1001	Inlet Safety Valve-BBOU Vent Line	Field	Safety Valve	-
76	BBOU-PSV-1002	Inlet Safety Valve-BBOU Vent Line	Field	Safety Valve	-
77	BBOU-AE/AT-1001	Point Type Gas Detector	Field	IR Type HC Detector	AI
78	BBOU-AI-1001	Point Type Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
79	BBOU-AI-1001A	Point Type Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
80	BBOU-AAH-1001	Point Type Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft

81	BBOU-AAHH-1001	Point Type Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
82	BBOU-AAH-1001A	Point Type Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
83	BBOU-AAHH-1001A	Point Type Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
84	BBOU-AE/AT-1002	Point Type Gas Detector	Field	IR Type HC Detector	AI
85	BBOU-AI-1002	Point Type Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
86	BBOU-AI-1002A	Point Type Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
87	BBOU-AAH-1002	Point Type Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
88	BBOU-AAHH-1002	Point Type Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
89	BBOU-AAH-1002A	Point Type Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
90	BBOU-AAHH-1002A	Point Type Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
91	BBOU-AE/AT-1003T	Open Path Gas Detector (Transmitter)	Field	IR Type Open Path HC Detector-Transmit	24V Power
92	BBOU-AE/AT-1003R	Open Path Gas Detector (Receiver)	Field	IR Type Open Path HC Detector-Receiver	AI
93	BBOU-AI-1003	Open Path Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
94	BBOU-AI-1003A	Open Path Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
95	BBOU-AAH-1003	Open Path Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
96	BBOU-AAHH-1003	Open Path Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
97	BBOU-AAH-1003A	Open Path Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
98	BBOU-AAHH-1003A	Open Path Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
99	BBOU-AE/AT-1004T	Open Path Gas Detector (Transmitter)	Field	IR Type Open Path HC Detector-Transmit	24V Power
100	BBOU-AE/AT-1004R	Open Path Gas Detector (Receiver)	Field	IR Type Open Path HC Detector-Receiver	AI
101	BBOU-AI-1004	Open Path Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
102	BBOU-AI-1004A	Open Path Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
103	BBOU-AAH-1004	Open Path Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
104	BBOU-AAHH-1004	Open Path Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
105	BBOU-AAH-1004A	Open Path Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
106	BBOU-AAHH-1004A	Open Path Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
107	BBOU-MAC-1001A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
108	BBOU-MAA-1001A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
109	BBOU-MAA-1001B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
110	BBOU-MAC-1002A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
111	BBOU-MAA-1002A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
112	BBOU-MAA-1002B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
113	BBOU-MAC-1003A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
114	BBOU-MAA-1003A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
115	BBOU-MAA-1003B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
116	BBOU-MAC-1004A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
117	BBOU-MAA-1004A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
118	BBOU-MAA-1004B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
119	BBOU-HB-1004	Hooter and beacon combine	Field	Hooter-Beacon	DO
120	BBOU-HBI-1004A	Hooter-Beacon Display at RTU	PLC-RTU	On PLC-RTU Screen	Soft
121	BBOU-HBI-1004B	Hooter Beacon Display at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial

122	BBOU-PI-1006	Battery Limit Pressure at the Inlet of Pig Receiver BBOU-	Field	Pressure Gauge	-
123	BBOU-PIT-1006	Battery Limit Pressure at the Inlet of Pig Receiver BBOU-	Field	Pressure Transmitter	AI
124	BBOU-PI-1006A	Battery Limit Pressure indication at RTU-Pig Receiver BBOU-	PLC-RTU	On PLC-RTU Screen	Soft
125	BBOU-PI-1006B	Battery Limit Pressure indication at remote DCS-Pig Receiver BBOU-	Remote DCS	On Remote DCS Screen	V-SAT/Serial
126	BBOU-TW/TI-1002	Battery Limit Temperature-Pig Receiver BBOU-A-100- Future	Field	Temperature Gauge with Thermowell	-
127	BBOU-TW/TE-1002	Battery Limit Temperature-Pig Receiver BBOU-A-100- Future	Field	RTD with Thermowell	-
128	BBOU-TIT-1002	Battery Limit Temperature-Pig Receiver BBOU-A-100- Future	Field	Temperature Transmitter	AI
129	BBOU-TI-1002A	Battery Limit Temperature at RTU-Pig Receiver BBOU-A-100	PLC-RTU	On PLC-RTU Screen	Soft
130	BBOU-TI-1002B	Battery Limit Temperature at remote DCS-Pig Receiver BBOU-A-100	DCS	On Remote DCS Screen	V-SAT/Serial
131	BBOU-SDV-1002	Inlet Shutdown Valve-OPUGBENE Pig Receiver BBOU-A-100	Field	Shutdown Valve- Gas Over Liquid	-
132	BBOU-SD-1002	ESD Command from RTU- Future	PLC-RTU	Push Button	DI
133	BBOU-SD-1002A	ESD Command from Remote DCS- Future	DCS	Push Button	V-SAT/Serial
134	BBOU-SOV-1002A	Inlet SDV Close Signal- Future	Field	Solenoid Valve	DO
135	BBOU-SOV-1002B	Inlet SDV Open Signal- Future	Field	Solenoid Valve	DO
136	BBOU-ZSC-1002	Inlet SDV Close Feedback Limit Switch- Future	Field	Limit Switch- Close	DI
137	BBOU-ZSO-1002	Inlet SDV Open Feedback Limit Switch- Future	Field	Limit Switch-Open	DI
138	BBOU-ZIC-1002	Inlet SDV Close Feedback at PLC-RTU- Future	PLC-RTU	On PLC-RTU Screen	Soft
139	BBOU-ZIO-1002	Inlet SDV Open Feedback at PLC-RTU- Future	PLC-RTU	On PLC-RTU Screen	Soft
140	BBOU-ZIC-1002A	Inlet SDV Close Feedback at remote DCS- Future	DCS	On Remote DCS Screen	V-SAT/Serial
141	BBOU-ZIO-1002A	Inlet SDV Open Feedback at remote DCS- Future	DCS	On Remote DCS Screen	V-SAT/Serial
142	BBOU-ZI-1001	Pig Indicator at the Inlet of the Pig Receiver BBOU-A-100	Field	Local Pig Signaller	-
143	BBOU-ZI-1002	Pig Indicator on the Pig Receiver BBOU-A-100 Future	Field	Local Pig Signaller	-
144	BBOU-PI-1007	Pressure at the Pig Receiver BBOU-A-100 Future	Field	Pressure Gauge	-
145	BBOU-PI-1008	Pressure on the Pig Receiver BBOU-A-100 Future	Field	Pressure Gauge	-
146	BBOU-PI-1009	Pressure on the Pig Receiver BBOU-A-100 Future	Field	Pressure Gauge	-
147	BBOU-PSV-1002	Pressure Safety Valve to Vent BBOU-A-100 Future	Field	Safety Valve	-
148	BBOU-RO-1002	Restriction Orifice to Vent - BBOU-A-100 Future	Field	Restriction Orifice	-
149	BBOU-SDV-1003	Outlet Shutdown Valve- Pig Launcher BBOU-A-200	Field	Shutdown Valve- Gas Over Liquid	-
150	BBOU-SD-1003	ESD Command from RTU	PLC-RTU	Push Button	DI
151	BBOU-SD-1003A	ESD Command from Remote DCS	DCS	Push Button	V-SAT/Serial
152	BBOU-SOV-1003A	Outlet SDV Close Signal	Field	Solenoid Valve	DO
153	BBOU-SOV-1003B	Outlet SDV Open Signal	Field	Solenoid Valve	DO
154	BBOU-ZSC-1003	Outlet SDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
155	BBOU-ZSO-1003	Outlet SDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
156	BBOU-ZIC-1003	Outlet SDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
157	BBOU-ZIO-1003	Outlet SDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
158	BBOU-ZIC-1003A	Outlet SDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
159	BBOU-ZIO-1003A	Outlet SDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
160	BBOU-ZI-1003	Pig Indicator at the outlet of the Pig Launcher BBOU-A-200	Field	Local Pig Signaller	-
161	BBOU-ZI-1004	Pig Indicator on the Pig Launcher BBOU-A-200	Field	Local Pig Signaller	-
162	BBOU-PI-1010	Pressure on the Pig Launcher BBOU-A-200	Field	Pressure Gauge	-

163	BBOU-PI-1011	Pressure on the Pig Launcher BBOU-A-200	Field	Pressure Gauge	-
164	BBOU-PI-1012	Pressure at the outlet of the Pig Launcher BBOU-A-200	Field	Pressure Gauge	-
165	BBOU-PSV-1004	Pressure Safety Valve to Vent BBOU-A-200	Field	Safety Valve	-
166	BBOU-RO-1003	Restriction Orifice to Vent - BBOU-A-200	Field	Restriction Orifice	-
167	BBOU-PIT-1013	Flowline Pressure for interlock-BBOU-A-200 Launcher	Field	Pressure Transmitter	AI
168	BBOU-PI-1013A	Flowline Pressure Indication at RTU- BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
169	BBOU-PI-1013B	Flowline Pressure Indication at remote DCS -BBOU-A-200	Remote DCS	On Remote DCS Screen	V-SAT/Serial
170	BBOU-PAL-1013	Flowline Pressure Low Alarm at RTU -BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
171	BBOU-PAH-1013	Flowline Pressure High Alarm at RTU - BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
172	BBOU-PAL-1013A	Flowline Pressure Low Alarm at remote DCS -BBOU-A-200	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
173	BBOU-PAH-1013A	Flowline Pressure High Alarm at remote DCS - BBOU-A-200	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
174	BBOU-PIT-1014	Flowline Pressure for interlock-BBOU-A-200 Launcher	Field	Pressure Transmitter	AI
175	BBOU-PI-1014A	Flowline Pressure Indication at RTU- BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
176	BBOU-PI-1014B	Flowline Pressure Indication at remote DCS -BBOU-A-200	Remote DCS	On Remote DCS Screen	V-SAT/Serial
177	BBOU-PAL-1014	Flowline Pressure Low Alarm at RTU -BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
178	BBOU-PAH-1014	Flowline Pressure High Alarm at RTU - BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
179	BBOU-PAL-1014A	Flowline Pressure Low Alarm at remote DCS -BBOU-A-200	Remote DCS	On Remote DCS Screen	V-SAT/Serial
180	BBOU-PAH-1014A	Flowline Pressure High Alarm at remote DCS - BBOU-A-200	Remote DCS	On Remote DCS Screen	V-SAT/Serial
181	BBOU-PIT-1015	Flowline Pressure for interlock-BBOU-A-200 Launcher	Field	Pressure Transmitter	AI
182	BBOU-PI-1015A	Flowline Pressure Indication at RTU- BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
183	BBOU-PI-1015B	Flowline Pressure Indication at remote DCS -BBOU-A-200	Remote DCS	On Remote DCS Screen	V-SAT/Serial
184	BBOU-PAL-1015	Flowline Pressure Low Alarm at RTU -BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
185	BBOU-PAH-1015	Flowline Pressure High Alarm at RTU - BBOU-A-200 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
186	BBOU-PAL-1015A	Flowline Pressure Low Alarm at remote DCS -BBOU-A-200	Remote DCS	On Remote DCS Screen	V-SAT/Serial
187	BBOU-PAH-1015A	Flowline Pressure High Alarm at remote DCS - BBOU-A-200	Remote DCS	On Remote DCS Screen	V-SAT/Serial
188	BBOU-PIT-1016	Pressure measurement of closed Drain Vessel BBOU-CDT-001	Field	Pressure Transmitter	AI
189	BBOU-PI-1016A	Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
190	BBOU-PI-1016B	Pressure Indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
191	BBOU-LIT-1001	Level for interlock-Closed Drain Vessel BBOU-CDT-001	Field	Guided Wave Radar Transmitter	AI
192	BBOU-LI-1001A	Level Indication at RTU- Closed Drain Vessel BBOU-CDT-001	PLC-RTU	On PLC-RTU Screen	Soft
193	BBOU-LI-1001B	Level Indication at remote DCS -Closed Drain Vessel BBOU-CDT-001	Remote DCS	On Remote DCS Screen	V-SAT/Serial
194	BBOU-LAL-1001	Level Low Alarm at RTU -Closed Drain Vessel BBOU-CDT-001	PLC-RTU	On PLC-RTU Screen	Soft
195	BBOU-LAH-1001	Level High Alarm at RTU - Closed Drain Vessel BBOU-CDT-001	PLC-RTU	On PLC-RTU Screen	Soft
196	BBOU-LAL-1001A	Level Low Alarm at remote DCS -Closed Drain Vessel BBOU-CDT-001	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
197	BBOU-LAH-1001A	Level High Alarm at remote DCS - Closed Drain Vessel BBOU-CDT-001	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
198	BBOU-LG-1001	Level Indication at Closed Drain Vessel BBOU-CDT-001	Field	Level Gauge	-
199	BBOU-TW/TE-1003	Temperature on the Closed Drain Vessel BBOU-CDT-001	Field	RTD with Thermowell	-
200	BBOU-TIT-1003	Temperature on the Closed Drain Vessel BBOU-CDT-001	Field	Temperature Transmitter	AI
201	BBOU-TI-1003A	Temperature indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
202	BBOU-TI-1003B	Temperature indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
203	BBOU-LIT-1002	Level at BBOU-ODT-001	Field	Guided Wave Radar Transmitter	AI

204	BBOU-LI-1002A	Level Indication at RTU- BBOU-ODT-001	PLC-RTU	On PLC-RTU Screen	Soft
205	BBOU-LI-1002B	Level Indication at remote DCS -BBOU-ODT-001	Remote DCS	On Remote DCS Screen	V-SAT/Serial
206	BBOU-LDS-1001	Leak Detection System	PLC-RTU	On PLC-RTU Screen	Serial

Notes

1. This point data base is tentative, subject to the finalisation of P&ID.
2. Interface signals from SPDC are not captured in this document as there is no input available.
3. Interface Signals for Pipeline Leak Detections shall be detailed out during detail engineering
4. Communication interface and failure signals shall be detailed out during detail engineering.

Instrumentation Point Database-ELEP (Revision- R03)

Sr. No.	Tag No.	Application	Location	Type of Instruments	IO Type
1	ELEP-FE/FT-1001	Check Meter at the inlet manifold	Field	Venturi Meter	-
2	ELEP-DPT-1001	Differential Pressure Measurement at the inlet across V	Field	DP Transmitter	AI
3	ELEP-FI-1001A	Flow Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
4	ELEP-FI-1001B	Flow Indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
5	ELEP-FQI-1001A	Flow Totalizer at RTU	PLC-RTU	On PLC-RTU Screen	Soft
6	ELEP-FQI-1001B	Flow Totalizer at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
7	ELEP-FE/FT-1002	Check Meter at the inlet manifold	Field	Venturi Meter	-
8	ELEP-DPT-1002	Differential Pressure Measurement at the inlet across V	Field	DP Transmitter	AI
9	ELEP-FI-1002A	Flow Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
10	ELEP-FI-1002B	Flow Indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
11	ELEP-FQI-1002A	Flow Totalizer at RTU	PLC-RTU	On PLC-RTU Screen	Soft
12	ELEP-FQI-1002B	Flow Totalizer at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
13	ELEP-PI-1001	Battery Limit Pressure at the Inlet Manifold of ELEP	Field	Pressure Gauge	-
14	ELEP-PIT-1001	Battery Limit Pressure at the Inlet Manifold of ELEP	Field	Pressure Transmitter	AI
15	ELEP-PI-1001A	Battery Limit Pressure indication at RTU-Inlet Manifold	PLC-RTU	On PLC-RTU Screen	Soft
16	ELEP-PI-1001B	Battery Limit Pressure indication at remote DCS-Inlet M	Remote DCS	On Remote DCS Screen	V-SAT/Serial
17	ELEP-PAL-1001	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
18	ELEP-PAH-1001	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
19	ELEP-PAL-1001A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
20	ELEP-PAH-1001A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
21	ELEP-TW/TI-1001	Battery Limit Temperature at the Inlet Manifold of ELEP	Field	Temperature Gauge with Thermowell	-
22	ELEP-TW/TE-1001	Battery Limit Temperature at the Inlet Manifold of ELEP	Field	RTD with Thermowell	-
23	ELEP-TIT-1001	Battery Limit Temperature at the Inlet Manifold of ELEP	Field	Temperature Transmitter	AI
24	ELEP-TI-1001A	Battery Limit Temperature at RTU-Inlet Manifold of ELEP	PLC-RTU	On PLC-RTU Screen	Soft
25	ELEP-TI-1001B	Battery Limit Temperature at remote DCS-Inlet Manifo	DCS	On Remote DCS Screen	V-SAT/Serial
26	ELEP-PCV-1001	Inlet pressure reduction-ELEP Gathering Manifold	Field	Pressure Regulator	-
27	ELEP-SDV-1001	Inlet Shutdown Valve-ELEP Gathering Manifold	Field	Shutdown Valve- Gas Over Liquid	-
28	ELEP-SD-1001	ESD Command from RTU	PLC-RTU	Push Button	DI
29	ELEP-SD-1001A	ESD Command from Remote DCS	DCS	Push Button	V-SAT/Serial
30	ELEP-SOV-1001A	Inlet SDV Close Signal	Field	Solenoid Valve	DO
31	ELEP-SOV-1001B	Inlet SDV Open Signal	Field	Solenoid Valve	DO
32	ELEP-ZSC-1001	Inlet SDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
33	ELEP-ZSO-1001	Inlet SDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
34	ELEP-ZIC-1001	Inlet SDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
35	ELEP-ZIO-1001	Inlet SDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
36	ELEP-ZIC-1001A	Inlet SDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
37	ELEP-ZIO-1001A	Inlet SDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
38	ELEP-PI-1002	Flowline Pressure to ELEP-A-200 Launcher	Field	Pressure Gauge	-
39	ELEP-PIT-1002	Flowline Pressure for interlock-ELEP Gathering Manifo	Field	Pressure Transmitter	AI
40	ELEP-PI-1002A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
41	ELEP-PI-1002B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
42	ELEP-PAL-1002	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft

43	ELEP-PAH-1002	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
44	ELEP-PAL-1002A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
45	ELEP-PAH-1002A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
46	ELEP-PIT-1003	Flowline Pressure for interlock-ELEP Gathering Manifold	Field	Pressure Transmitter	AI
47	ELEP-PI-1003A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
48	ELEP-PI-1003B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
49	ELEP-PAL-1003	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
50	ELEP-PAH-1003	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
51	ELEP-PAL-1003A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
52	ELEP-PAH-1003A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
53	ELEP-PIT-1004	Flowline Pressure for interlock-ELEP Gathering Manifold	Field	Pressure Transmitter	AI
54	ELEP-PI-1004A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
55	ELEP-PI-1004B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
56	ELEP-PAL-1004	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
57	ELEP-PAH-1004	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
58	ELEP-PAL-1004A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
59	ELEP-PAH-1004A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
60	ELEP-BDV-1002	Blowdown Valve to Vent	Field	Blowdown Valve- Gas Over Liquid	-
61	ELEP-SD-1002	ESD Command from RTU	PLC-RTU	Push Button	DI
62	ELEP-SD-1002A	ESD Command from Remote DCS	DCS	Push Button	V-SAT/Serial
63	ELEP-SOV-1002A	BDV Close Signal	Field	Solenoid Valve	DO
64	ELEP-SOV-1002B	BDV Open Signal	Field	Solenoid Valve	DO
65	ELEP-ZSC-1002	BDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
66	ELEP-ZSO-1002	BDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
67	ELEP-ZIC-1002	BDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
68	ELEP-ZIO-1002	BDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
69	ELEP-ZIC-1002A	BDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
70	ELEP-ZIO-1002A	BDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
71	ELEP-PIT-1005	Instrument Gas Manifold Pressure Measurement	Field	Pressure Transmitter	AI
72	ELEP-PI-1005A	Instrument Gas Manifold Pressure Measurement- RTU Indication	PLC-RTU	On PLC-RTU Screen	Soft
73	ELEP-PI-1005B	Instrument Gas Manifold Pressure Measurement- Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
74	ELEP-RO-1001	Restriction Orifice to Vent-ELEP Vent Line Depressurisation	Field	Restriction Orifice	-
75	ELEP-PSV-1001	Inlet Safety Valve-ELEP Vent Line	Field	Safety Valve	-
76	ELEP-PSV-1002	Inlet Safety Valve-ELEP Vent Line	Field	Safety Valve	-
77	ELEP-AE/AT-1001	Point Type Gas Detector	Field	IR Type HC Detector	AI
78	ELEP-AI-1001	Point Type Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
79	ELEP-AI-1001A	Point Type Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
80	ELEP-AAH-1001	Point Type Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
81	ELEP-AAHH-1001	Point Type Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
82	ELEP-AAH-1001A	Point Type Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
83	ELEP-AAHH-1001A	Point Type Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
84	ELEP-AE/AT-1002	Point Type Gas Detector	Field	IR Type HC Detector	AI
85	ELEP-AI-1002	Point Type Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
86	ELEP-AI-1002A	Point Type Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial

87	ELEP-AAH-1002	Point Type Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
88	ELEP-AAHH-1002	Point Type Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
89	ELEP-AAH-1002A	Point Type Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
90	ELEP-AAHH-1002A	Point Type Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
91	ELEP-AE/AT-1003T	Open Path Gas Detector (Transmitter)	Field	IR Type Open Path HC Detector-Transmitter	24V Power
92	ELEP-AE/AT-1003R	Open Path Gas Detector (Receiver)	Field	IR Type Open Path HC Detector-Receiver	AI
93	ELEP-AI-1003	Open Path Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
94	ELEP-AI-1003A	Open Path Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
95	ELEP-AAH-1003	Open Path Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
96	ELEP-AAHH-1003	Open Path Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
97	ELEP-AAH-1003A	Open Path Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
98	ELEP-AAHH-1003A	Open Path Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
99	ELEP-AE/AT-1004T	Open Path Gas Detector (Transmitter)	Field	IR Type Open Path HC Detector-Transmitter	24V Power
100	ELEP-AE/AT-1004R	Open Path Gas Detector (Receiver)	Field	IR Type Open Path HC Detector-Receiver	AI
101	ELEP-AI-1004	Open Path Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
102	ELEP-AI-1004A	Open Path Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
103	ELEP-AAH-1004	Open Path Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
104	ELEP-AAHH-1004	Open Path Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
105	ELEP-AAH-1004A	Open Path Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
106	ELEP-AAHH-1004A	Open Path Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
107	ELEP-MAC-1001A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
108	ELEP-MAA-1001A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
109	ELEP-MAA-1001B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
110	ELEP-MAC-1002A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
111	ELEP-MAA-1002A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
112	ELEP-MAA-1002B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
113	ELEP-MAC-1003A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
114	ELEP-MAA-1003A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
115	ELEP-MAA-1003B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
116	ELEP-MAC-1004A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
117	ELEP-MAA-1004A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
118	ELEP-MAA-1004B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
119	ELEP-HB-1004	Hooter and beacon combine	Field	Hooter-Beacon	DO
120	ELEP-HBI-1004A	Hooter-Beacon Display at RTU	PLC-RTU	On PLC-RTU Screen	Soft
121	ELEP-HBI-1004B	Hooter Beacon Display at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
122	ELEP-PI-1006	Battery Limit Pressure at the Inlet of Pig Receiver ELEP-	Field	Pressure Gauge	-
123	ELEP-PIT-1006	Battery Limit Pressure at the Inlet of Pig Receiver ELEP-	Field	Pressure Transmitter	AI
124	ELEP-PI-1006A	Battery Limit Pressure indication at RTU-Pig Receiver ELEP-	PLC-RTU	On PLC-RTU Screen	Soft
125	ELEP-PI-1006B	Battery Limit Pressure indication at remote DCS-Pig Receiver ELEP-	Remote DCS	On Remote DCS Screen	V-SAT/Serial
126	ELEP-TW/TI-1002	Battery Limit Temperature-Pig Receiver ELEP-A-100- Full	Field	Temperature Gauge with Thermowell	-
127	ELEP-TW/TE-1002	Battery Limit Temperature-Pig Receiver ELEP-A-100- Full	Field	RTD with Thermowell	-
128	ELEP-TIT-1002	Battery Limit Temperature-Pig Receiver ELEP-A-100- Full	Field	Temperature Transmitter	AI
129	ELEP-TI-1002A	Battery Limit Temperature at RTU-Pig Receiver ELEP-A-100- Full	PLC-RTU	On PLC-RTU Screen	Soft
130	ELEP-TI-1002B	Battery Limit Temperature at remote DCS-Pig Receiver ELEP-A-100- Full	DCS	On Remote DCS Screen	V-SAT/Serial

131	ELEP-SDV-1002	Inlet Shutdown Valve-OPUGBENE Pig Receiver ELEP-A-1	Field	Shutdown Valve- Gas Over Liquid	-
132	ELEP-SD-1002	ESD Command from RTU- Future	PLC-RTU	Push Button	DI
133	ELEP-SD-1002A	ESD Command from Remote DCS- Future	DCS	Push Button	V-SAT/Serial
134	ELEP-SOV-1002A	Inlet SDV Close Signal- Future	Field	Solenoid Valve	DO
135	ELEP-SOV-1002B	Inlet SDV Open Signal- Future	Field	Solenoid Valve	DO
136	ELEP-ZSC-1002	Inlet SDV Close Feedback Limit Switch- Future	Field	Limit Switch- Close	DI
137	ELEP-ZSO-1002	Inlet SDV Open Feedback Limit Switch- Future	Field	Limit Switch-Open	DI
138	ELEP-ZIC-1002	Inlet SDV Close Feedback at PLC-RTU- Future	PLC-RTU	On PLC-RTU Screen	Soft
139	ELEP-ZIO-1002	Inlet SDV Open Feedback at PLC-RTU- Future	PLC-RTU	On PLC-RTU Screen	Soft
140	ELEP-ZIC-1002A	Inlet SDV Close Feedback at remote DCS- Future	DCS	On Remote DCS Screen	V-SAT/Serial
141	ELEP-ZIO-1002A	Inlet SDV Open Feedback at remote DCS- Future	DCS	On Remote DCS Screen	V-SAT/Serial
142	ELEP-ZI-1001	Pig Indicator at the Inlet of the Pig Receiver ELEP-A-100	Field	Local Pig Signaller	-
143	ELEP-ZI-1002	Pig Indicator on the Pig Receiver ELEP-A-100 Future	Field	Local Pig Signaller	-
144	ELEP-PI-1007	Pressure at the Pig Receiver ELEP-A-100 Future	Field	Pressure Gauge	-
145	ELEP-PI-1008	Pressure on the Pig Receiver ELEP-A-100 Future	Field	Pressure Gauge	-
146	ELEP-PI-1009	Pressure on the Pig Receiver ELEP-A-100 Future	Field	Pressure Gauge	-
147	ELEP-PSV-1003	Pressure Safety Valve to Vent ELEP-A-100 Future	Field	Safety Valve	-
148	ELEP-RO-1002	Restriction Orifice to Vent - ELEP-A-100 Future	Field	Restriction Orifice	-
149	ELEP-PI-1010	Battery Limit Pressure at the Inlet of Pig Receiver ELEP-	Field	Pressure Gauge	-
150	ELEP-PIT-1010	Battery Limit Pressure at the Inlet of Pig Receiver ELEP-	Field	Pressure Transmitter	AI
151	ELEP-PI-1010A	Battery Limit Pressure indication at RTU-Pig Receiver El	PLC-RTU	On PLC-RTU Screen	Soft
152	ELEP-PI-1010B	Battery Limit Pressure indication at remote DCS-Pig Re	Remote DCS	On Remote DCS Screen	V-SAT/Serial
153	ELEP-TW/TI-1003	Battery Limit Temperature-Pig Receiver ELEP-A-200	Field	Temperature Gauge with Thermowell	-
154	ELEP-TW/TE-1003	Battery Limit Temperature-Pig Receiver ELEP-A-200	Field	RTD with Thermowell	-
155	ELEP-TIT-1003	Battery Limit Temperature-Pig Receiver ELEP-A-200	Field	Temperature Transmitter	AI
156	ELEP-TI-1003A	Battery Limit Temperature at RTU-Pig Receiver ELEP-A-	PLC-RTU	On PLC-RTU Screen	Soft
157	ELEP-TI-1003B	Battery Limit Temperature at remote DCS-Pig Receiver	DCS	On Remote DCS Screen	V-SAT/Serial
158	ELEP-SDV-1003	Inlet Shutdown Valve-OPUGBENE Pig Receiver ELEP-A-2	Field	Shutdown Valve- Gas Over Liquid	-
159	ELEP-SD-1003	ESD Command from RTU	PLC-RTU	Push Button	DI
160	ELEP-SD-1003A	ESD Command from Remote DCS	DCS	Push Button	V-SAT/Serial
161	ELEP-SOV-1003A	Inlet SDV Close Signal	Field	Solenoid Valve	DO
162	ELEP-SOV-1003B	Inlet SDV Open Signal	Field	Solenoid Valve	DO
163	ELEP-ZSC-1003	Inlet SDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
164	ELEP-ZSO-1003	Inlet SDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
165	ELEP-ZIC-1003	Inlet SDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
166	ELEP-ZIO-1003	Inlet SDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
167	ELEP-ZIC-1003A	Inlet SDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
168	ELEP-ZIO-1003A	Inlet SDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
169	ELEP-ZI-1003	Pig Indicator at the Inlet of the Pig Receiver ELEP-A-200	Field	Local Pig Signaller	-
170	ELEP-ZI-1004	Pig Indicator on the Pig Receiver ELEP-A-200	Field	Local Pig Signaller	-
171	ELEP-PI-1011	Pressure at the Pig Receiver ELEP-A-200	Field	Pressure Gauge	-
172	ELEP-PI-1012	Pressure on the Pig Receiver ELEP-A-200	Field	Pressure Gauge	-
173	ELEP-PI-1013	Pressure on the Pig Receiver ELEP-A-200	Field	Pressure Gauge	-
174	ELEP-PSV-1004	Pressure Safety Valve to Vent ELEP-A-200	Field	Safety Valve	-

175	ELEP-RO-1003	Restriction Orifice to Vent - ELEP-A-200	Field	Restriction Orifice	-
176	ELEP-AE/AT-1005	Point Type Gas Detector	Field	IR Type HC Detector	AI
177	ELEP-AI-1005	Point Type Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
178	ELEP-AI-1005A	Point Type Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
179	ELEP-AAH-1005	Point Type Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
180	ELEP-AAHH-1005	Point Type Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
181	ELEP-AAH-1005A	Point Type Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
182	ELEP-AAHH-1005A	Point Type Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
183	ELEP-AE/AT-1006T	Open Path Gas Detector (Transmitter)	Field	IR Type Open Path HC Detector-Transmitter	24V Power
184	ELEP-AE/AT-1006R	Open Path Gas Detector (Receiver)	Field	IR Type Open Path HC Detector-Receiver	AI
185	ELEP-AI-1006	Open Path Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
186	ELEP-AI-1006A	Open Path Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
187	ELEP-AAH-1006	Open Path Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
188	ELEP-AAHH-1006	Open Path Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
189	ELEP-AAH-1006A	Open Path Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
190	ELEP-AAHH-1006A	Open Path Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
191	ELEP-MAC-1005A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
192	ELEP-MAA-1005A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
193	ELEP-MAA-1005B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
194	ELEP-SDV-1004	Outlet Shutdown Valve- Pig Launcher ELEP-A-300	Field	Shutdown Valve- Gas Over Liquid	-
195	ELEP-SD-1004	ESD Command from RTU	PLC-RTU	Push Button	DI
196	ELEP-SD-1004A	ESD Command from Remote DCS	DCS	Push Button	V-SAT/Serial
197	ELEP-SOV-1004A	Outlet SDV Close Signal	Field	Solenoid Valve	DO
198	ELEP-SOV-1004B	Outlet SDV Open Signal	Field	Solenoid Valve	DO
199	ELEP-ZSC-1004	Outlet SDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
200	ELEP-ZSO-1004	Outlet SDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
201	ELEP-ZIC-1004	Outlet SDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
202	ELEP-ZIO-1004	Outlet SDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
203	ELEP-ZIC-1004A	Outlet SDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
204	ELEP-ZIO-1004A	Outlet SDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	V-SAT/Serial
205	ELEP-ZI-1005	Pig Indicator at the outlet of the Pig Launcher ELEP-A-300	Field	Local Pig Signaller	-
206	ELEP-ZI-1006	Pig Indicator on the Pig Launcher ELEP-A-300	Field	Local Pig Signaller	-
207	ELEP-PI-1014	Pressure on the Pig Launcher ELEP-A-300	Field	Pressure Gauge	-
208	ELEP-PI-1015	Pressure on the Pig Launcher ELEP-A-300	Field	Pressure Gauge	-
209	ELEP-PI-1016	Pressure at the outlet of the Pig Launcher ELEP-A-300	Field	Pressure Gauge	-
210	ELEP-PSV-1005	Pressure Safety Valve to Vent ELEP-A-300	Field	Safety Valve	-
211	ELEP-RO-1004	Restriction Orifice to Vent - ELEP-A-300	Field	Restriction Orifice	-
212	ELEP-PIT-1018	Flowline Pressure for interlock-ELEP-A-300 Launcher	Field	Pressure Transmitter	AI
213	ELEP-PI-1018A	Flowline Pressure Indication at RTU- ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft

214	ELEP-PI-1018B	Flowline Pressure Indication at remote DCS -ELEP-A-300	Remote DCS	On Remote DCS Screen	V-SAT/Serial
215	ELEP-PAL-1018	Flowline Pressure Low Alarm at RTU -ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
216	ELEP-PAH-1018	Flowline Pressure High Alarm at RTU - ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
217	ELEP-PAL-1018A	Flowline Pressure Low Alarm at remote DCS -ELEP-A-300	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
218	ELEP-PAH-1018A	Flowline Pressure High Alarm at remote DCS - ELEP-A-300	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
219	ELEP-PIT-1019	Flowline Pressure for interlock-ELEP-A-300 Launcher	Field	Pressure Transmitter	AI
220	ELEP-PI-1019A	Flowline Pressure Indication at RTU- ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
221	ELEP-PI-1019B	Flowline Pressure Indication at remote DCS -ELEP-A-300	Remote DCS	On Remote DCS Screen	V-SAT/Serial
222	ELEP-PAL-1019	Flowline Pressure Low Alarm at RTU -ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
223	ELEP-PAH-1019	Flowline Pressure High Alarm at RTU - ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
224	ELEP-PAL-1019A	Flowline Pressure Low Alarm at remote DCS -ELEP-A-300	Remote DCS	On Remote DCS Screen	V-SAT/Serial
225	ELEP-PAH-1019A	Flowline Pressure High Alarm at remote DCS - ELEP-A-300	Remote DCS	On Remote DCS Screen	V-SAT/Serial
226	ELEP-PIT-1020	Flowline Pressure for interlock-ELEP-A-300 Launcher	Field	Pressure Transmitter	AI
227	ELEP-PI-1020A	Flowline Pressure Indication at RTU- ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
228	ELEP-PI-1020B	Flowline Pressure Indication at remote DCS -ELEP-A-300	Remote DCS	On Remote DCS Screen	V-SAT/Serial
229	ELEP-PAL-1020	Flowline Pressure Low Alarm at RTU -ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
230	ELEP-PAH-1020	Flowline Pressure High Alarm at RTU - ELEP-A-300 Launcher	PLC-RTU	On PLC-RTU Screen	Soft
231	ELEP-PAL-1020A	Flowline Pressure Low Alarm at remote DCS -ELEP-A-300	Remote DCS	On Remote DCS Screen	V-SAT/Serial
232	ELEP-PAH-1020A	Flowline Pressure High Alarm at remote DCS - ELEP-A-300	Remote DCS	On Remote DCS Screen	V-SAT/Serial
233	ELEP-PIT-1021	Pressure measurement of closed Drain Vessel ELEP-CDT-001	Field	Pressure Transmitter	AI
234	ELEP-PI-1021A	Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
235	ELEP-PI-1021B	Pressure Indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
236	ELEP-LIT-1001	Level for interlock-Closed Drain Vessel ELEP-CDT-001	Field	Guided Wave Radar Transmitter	AI
237	ELEP-LI-1001A	Level Indication at RTU- Closed Drain Vessel ELEP-CDT-001	PLC-RTU	On PLC-RTU Screen	Soft
238	ELEP-LI-1001B	Level Indication at remote DCS -Closed Drain Vessel ELEP-CDT-001	Remote DCS	On Remote DCS Screen	V-SAT/Serial
239	ELEP-LAL-1001	Level Low Alarm at RTU -Closed Drain Vessel ELEP-CDT-001	PLC-RTU	On PLC-RTU Screen	Soft
240	ELEP-LAH-1001	Level High Alarm at RTU - Closed Drain Vessel ELEP-CDT-001	PLC-RTU	On PLC-RTU Screen	Soft
241	ELEP-LAL-1001A	Level Low Alarm at remote DCS -Closed Drain Vessel ELEP-CDT-001	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
242	ELEP-LAH-1001A	Level High Alarm at remote DCS - Closed Drain Vessel ELEP-CDT-001	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
243	ELEP-LG-1001	Level Indication at Closed Drain Vessel ELEP-CDT-001	Field	Level Gauge	-
244	ELEP-TW/TE-1003	Temperature on the Closed Drain Vessel ELEP-CDT-001	Field	RTD with Thermowell	-
245	ELEP-TIT-1004	Temperature on the Closed Drain Vessel ELEP-CDT-001	Field	Temperature Transmitter	AI
246	ELEP-TI-1004A	Temperature indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
247	ELEP-TI-1004B	Temperature indication at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
248	ELEP-LIT-1002	Level at ELEP-ODT-001	Field	Guided Wave Radar Transmitter	AI
249	ELEP-LI-1002A	Level Indication at RTU- ELEP-ODT-001	PLC-RTU	On PLC-RTU Screen	Soft
250	ELEP-LI-1002B	Level Indication at remote DCS -ELEP-ODT-001	Remote DCS	On Remote DCS Screen	V-SAT/Serial
251	ELEP-LDS-1001	Leak Detection System	PLC-RTU	On PLC-RTU Screen	Serial

Notes

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4. Communication interface and failure signals shall be detailed out during detail engineering.

Instrumentation Point Database-AKAB (Revision- R03)

Sr. No.	Tag No.	Application	Location	Type of Instruments	IO Type
1	AKAB-PI-1001	Battery Limit Pressure at the Inlet of Pig Receiver AKAB-A-100	Field	Pressure Gauge	-
2	AKAB-PIT-1001	Battery Limit Pressure at the Inlet of Pig Receiver AKAB-A-100	Field	Pressure Transmitter	AI
3	AKAB-PI-1001A	Battery Limit Pressure indication at RTU-Pig Receiver AKAB-A-100	PLC-RTU	On PLC-RTU Screen	Soft
4	AKAB-PI-1001B	Battery Limit Pressure indication at remote DCS-Pig Receiver AKAB-A-100	Remote DCS	On Remote DCS Screen	Serial
5	AKAB-TW/TI-1001	Battery Limit Temperature-Pig Receiver AKAB-A-100	Field	Temperature Gauge with Thermowell	-
6	AKAB-TW/TE-1001	Battery Limit Temperature-Pig Receiver AKAB-A-100	Field	RTD with Thermowell	-
7	AKAB-TIT-1001	Battery Limit Temperature-Pig Receiver AKAB-A-100	Field	Temperature Transmitter	AI
8	AKAB-TI-1001A	Battery Limit Temperature at RTU-Pig Receiver AKAB-A-100	PLC-RTU	On PLC-RTU Screen	Soft
9	AKAB-TI-1001B	Battery Limit Temperature at remote DCS-Pig Receiver AKAB-A-100	DCS	On Remote DCS Screen	Serial
10	AKAB-SDV-1001	Inlet Shutdown Valve-OPUGBENE Pig Receiver AKAB-A-100	Field	Shutdown Valve- Gas Over Liquid	-
11	AKAB-SD-1001	ESD Command from RTU	PLC-RTU	Push Button	DI
12	AKAB-SD-1001A	ESD Command from Remote DCS	DCS	Push Button	Serial
13	AKAB-SOV-1001A	Inlet SDV Close Signal	Field	Solenoid Valve	DO
14	AKAB-SOV-1001B	Inlet SDV Open Signal	Field	Solenoid Valve	DO
15	AKAB-ZSC-1001	Inlet SDV Close Feedback Limit Switch	Field	Limit Switch- Close	DI
16	AKAB-ZSO-1001	Inlet SDV Open Feedback Limit Switch	Field	Limit Switch-Open	DI
17	AKAB-ZIC-1001	Inlet SDV Close Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
18	AKAB-ZIO-1001	Inlet SDV Open Feedback at PLC-RTU	PLC-RTU	On PLC-RTU Screen	Soft
19	AKAB-ZIC-1001A	Inlet SDV Close Feedback at remote DCS	DCS	On Remote DCS Screen	Serial
20	AKAB-ZIO-1001A	Inlet SDV Open Feedback at remote DCS	DCS	On Remote DCS Screen	Serial
21	AKAB-PIT-1002	Flowline Pressure for interlock-AKAB Gathering Manifold	Field	Pressure Transmitter	AI
22	AKAB-PI-1002A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
23	AKAB-PI-1002B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
24	AKAB-PAL-1002	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
25	AKAB-PAH-1002	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
26	AKAB-PAL-1002A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
27	AKAB-PAH-1002A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
28	AKAB-PIT-1003	Flowline Pressure for interlock-AKAB Gathering Manifold	Field	Pressure Transmitter	AI
29	AKAB-PI-1003A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
30	AKAB-PI-1003B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
31	AKAB-PAL-1003	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
32	AKAB-PAH-1003	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
33	AKAB-PAL-1003A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
34	AKAB-PAH-1003A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
35	AKAB-PIT-1004	Flowline Pressure for interlock-AKAB Gathering Manifold	Field	Pressure Transmitter	AI
36	AKAB-PI-1004A	Flowline Pressure Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
37	AKAB-PI-1004B	Flowline Pressure Indication at remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
38	AKAB-PAL-1004	Flowline Pressure Low Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
39	AKAB-PAH-1004	Flowline Pressure High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
40	AKAB-PAL-1004A	Flowline Pressure Low Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial
41	AKAB-PAH-1004A	Flowline Pressure High Alarm at remote DCS	Remote DCS	On PLC-RTU Screen	V-SAT/Serial

42	AKAB-ZI-1003	Pig Indicator at the Inlet of the Pig Receiver AKAB-A-100	Field	Local Pig Signaller	-
43	AKAB-ZI-1004	Pig Indicator on the Pig Receiver AKAB-A-100	Field	Local Pig Signaller	-
44	AKAB-PI-1011	Pressure at the Pig Receiver AKAB-A-100	Field	Pressure Gauge	-
45	AKAB-PI-1012	Pressure on the Pig Receiver AKAB-A-100	Field	Pressure Gauge	-
46	AKAB-PI-1013	Pressure on the Pig Receiver AKAB-A-100	Field	Pressure Gauge	-
47	AKAB-PSV-1004	Pressure Safety Valve to Vent AKAB-A-100	Field	Safety Valve	-
48	AKAB-RO-1003	Restriction Orifice to Vent - AKAB-A-100	Field	Restriction Orifice	-
49	AKAB-AE/AT-1001	Point Type Gas Detector	Field	IR Type HC Detector	AI
50	AKAB-AI-1001	Point Type Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
51	AKAB-AI-1001A	Point Type Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	Radio/Serial
52	AKAB-AAH-1001	Point Type Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
53	AKAB-AAHH-1001	Point Type Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
54	AKAB-AAH-1001A	Point Type Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	Radio/Serial
55	AKAB-AAHH-1001A	Point Type Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	Radio/Serial
56	AKAB-AE/AT-1002T	Open Path Gas Detector (Transmitter)	Field	IR Type Open Path HC Detector-Transmitter	24V Power
57	AKAB-AE/AT-1002R	Open Path Gas Detector (Receiver)	Field	IR Type Open Path HC Detector-Receiver	AI
58	AKAB-AI-1002	Open Path Gas Detector Indication at RTU	PLC-RTU	On PLC-RTU Screen	Soft
59	AKAB-AI-1002A	Open Path Gas Detector Indication at remote DCS	Remote DCS	On Remote DCS Screen	Radio/Serial
60	AKAB-AAH-1002	Open Path Gas Detector High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
61	AKAB-AAHH-1002	Open Path Gas Detector High High Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
62	AKAB-AAH-1002A	Open Path Gas Detector High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	Radio/Serial
63	AKAB-AAHH-1002A	Open Path Gas Detector High High Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	Radio/Serial
64	AKAB-MAC-1001A	Manual Alarm Call Point (Field ESD)	Field	Manual Call Point	DI
65	AKAB-MAA-1001A	Manual Alarm Call Point Alarm at RTU	PLC-RTU	On PLC-RTU Screen	Soft
66	AKAB-MAA-1001B	Manual Alarm Call Point Alarm at Remote DCS	Remote DCS	On Remote DCS Screen	Radio/Serial
67	ELEP-HB-1003	Hooter and beacon combine	Field	Hooter-Beacon	DO
68	ELEP-HBI-1003A	Hooter-Beacon Display at RTU	PLC-RTU	On PLC-RTU Screen	Soft
69	ELEP-HBI-1003B	Hooter Beacon Display at Remote DCS	Remote DCS	On Remote DCS Screen	V-SAT/Serial
70	AKAB-LDS-1001	Leak Detection System	PLC-RTU	On PLC-RTU Screen	Serial

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